## Investigation: FIBAM CIA INDUSTRIAL (CVM code 6017), fiscal year 2015, as it was on 2018-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

## Company

Sector: Metalurgia e Siderurgia (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

## Document used

2015 DFP, version 1, filed on 2016-03-29; origin: open data; type: individual; original scale: MIL.
Versions known on the date: 1 (2016-03-29).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=54553

## Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: component not computable: DEPI
- Alert by the study 1 rule: does not trigger alerts
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: —
- Percentile in group and year: —
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: low

DSRI, days' sales in receivables index
- Value: 0.836 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 12 (n = 20)
- Percentile in group and year: 21 (n = 459)
- The company's own history: median 0.957; Tukey range (1.5 × IQR): from 0.876 to 1.067; value below the range (index; 5 fiscal years)
- Confidence in the data: medium

GMI, gross margin index
- Value: 6.089 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 98 (n = 20)
- Percentile in group and year: 99 (n = 480)
- The company's own history: median 1.099; Tukey range (1.5 × IQR): from 0.834 to 1.282; value above the range (index; 5 fiscal years)
- Confidence in the data: medium

AQI, asset quality index
- Value: 1.176 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 82 (n = 20)
- Percentile in group and year: 83 (n = 463)
- The company's own history: median 1.033; Tukey range (1.5 × IQR): from 0.916 to 1.119; value above the range (index; 5 fiscal years)
- Confidence in the data: medium

SGI, sales growth index

Value: 0.773 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 8 (n = 20)
Percentile in group and year: 15 (n = 482)
The company's own history: median 0.895; Tukey range (1.5 × IQR): from 0.758 to 1.111; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

DEPI, depreciation index
Value: missing account: depreciation, cash flow lines in section 6.01.01
M-score component; it has no rule of its own
Percentile in sector and year: —
Percentile in group and year: —
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: low

SGAI, SG&A expenses index
Value: 1.051 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 78 (n = 20)
Percentile in group and year: 59 (n = 469)
The company's own history: median 1.066; Tukey range (1.5 × IQR): from 0.879 to 1.231; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.146 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 42 (n = 20)
Percentile in group and year: 25 (n = 488)
The company's own history: median −0.063; Tukey range (1.5 × IQR): from −0.137 to 0.049; value below the range (index; 5 fiscal years)
Confidence in the data: medium

LVGI, leverage index
Value: 0.836 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 8 (n = 20)
Percentile in group and year: 8 (n = 483)
The company's own history: median 1.032; Tukey range (1.5 × IQR): from 0.619 to 1.468; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

DPO jump
Value: −39.0 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 42.5 days)
Percentile in sector and year: 3 (n = 19)
Percentile in group and year: 7 (n = 448)
The company's own history: median 1.3 days; Tukey range (1.5 × IQR): from −17.7 days to 24.4 days; value below the range (5 fiscal years)
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 10.59 p.p.
Alert by the study 1 rule: yes
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 52 (n = 20)
Percentile in group and year: 57 (n = 437)
The company's own history: median 6.73 p.p.; Tukey range (1.5 × IQR): from 5.02 p.p. to 8.06 p.p.; value above the range (5 fiscal years)
Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.134 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 42 (n = 20)
Percentile in group and year: 30 (n = 487)
The company's own history: median −0.062; Tukey range (1.5 × IQR): from −0.131 to 0.046; value below the range (index; 5 fiscal years)
Confidence in the data: medium

## Financial risk (context only)

Altman Z" (financial risk, context only)
Value: 0.360 (index)
Components: X1 = 0.228; X2 = −0.251; X3 = −0.062; X4 = 0.094 (indices)
Percentile in sector and year: 40 (n = 21)
The company's own history: median 1.044; Tukey range (1.5 × IQR): from −1.249 to 3.786; value inside the range (index; 5 fiscal years)
- No cut-off zone: the available source does not define zones for Z"
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

## Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 2/9 (8 computable)
Signals:
- Return on assets (ROA): 0
- Operating cash flow on assets (CFO): 0
- Change in ROA (ΔROA): 0
- Cash above earnings (ACCRUAL): 1
- Change in leverage (ΔLEVER): 0
- Change in current ratio (ΔLIQUID): 1
- No equity issuance (EQ_OFFER): —
- Change in gross margin (ΔMARGIN): 0
- Change in asset turnover (ΔTURN): 0
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

## Changes

Highlighted in the following year's comparative: 1 (threshold of R$0.7 million, 1% of total

assets of the version used). Values in R$ million:
DFC (indirect method) 6.01 Caixa Líquido Atividades Operacionais: −0.1 in the version used (v1, filed on 2016-03-29); −4.0 in the next DFP's comparative (v1, filed on 2017-03-27); difference of −3.9. Different value in the following year's comparative (restatement or reclassification).
Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets of the version used. The other differences (reclassifications between lines and smaller differences) stay in the full list.
Full list: 5 accounts above the R$1,000 tolerance (group a, totals and key lines: 1; group b, reclassifications: 4), in the annex at the end.

## What was known on the date

- DFP 2010, v1: 2011-03-29
- DFP 2010, v2: 2011-03-29
- DFP 2011, v1: 2012-03-14
- DFP 2011, v2: 2012-04-18
- DFP 2012, v1: 2013-03-12
- DFP 2013, v1: 2014-03-13
- DFP 2014, v1: 2015-03-25
- FCA 2015, v1: 2015-03-25
- FCA 2015, v2: 2015-04-29
- FCA 2015, v3: 2015-05-05
- DFP 2015, v1: 2016-03-29
- DFP 2016, v1: 2017-03-27

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

## Data quality

53 applicable checks; 0 failed.

## Auditor's report and auditor (factual section; no alert)

Auditor's report: Com Ressalva (qualified).
no opinion excerpt was found in the text
Auditor in the 2015 FCA: Apice Auditores Independentes S/S.
Auditor in the 2016 FCA: JPPS Auditores Independentes S/S.
Auditor in the 2017 FCA: JPPS Auditores Independentes S/S.
Auditor changes (one line per change, canonical name by the firm's CVM code):
- 2012: Padrão Auditoria S/S -> Narazzaqui & Cia. - Auditores. The outgoing firm audited 8 consecutive fiscal years (2004 to 2011), by the start and end dates in the FCA.
- 2013: Narazzaqui & Cia. - Auditores -> Binah Sp Auditores Independentes (transition: two firms in the 2013 FCA). The outgoing firm audited 1 consecutive fiscal years (2012 to 2012), by the start and end dates in the FCA.
- 2015: Binah Sp Auditores Independentes -> Apice Auditores Independentes S/S. The outgoing firm audited 2 consecutive fiscal years (2013 to 2014), by the start and end dates in the FCA.
- 2016: Apice Auditores Independentes S/S -> JPPS Auditores Independentes S/S. The outgoing firm audited 1 consecutive fiscal years (2015 to 2015), by the start and end dates in the FCA.

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

## How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

## Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

## Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 36 DFP lines, each with document, version, filing date, account and location.

## Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E). Generated at 2026-11-10T19:36:12+00:00.

## Today's information (outside the reference date)

Registry status: CANCELADA. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.

## Annex: full list of differences in the following year's comparative (R$ million)

(a) DFC (indirect method) 6.01 Caixa Líquido Atividades Operacionais: −0.1 in the version used, −4.0 in the comparative (difference of −3.9)
(b) DFC (indirect method) 6.05.01 Saldo Inicial de Caixa e Equivalentes: 1.5 in the version used, 7.9 in the comparative (difference of 6.3)
(b) DFC (indirect method) 6.01.01 Caixa Gerado nas Operações: −0.0 in the version used, −3.9 in the comparative (difference of −3.9)
(b) DFC (indirect method) 6.05 Aumento (Redução) de Caixa e Equivalentes: −0.3 in the version used, −4.2 in the comparative (difference of −3.9)
(b) DFC (indirect method) 6.05.02 Saldo Final de Caixa e Equivalentes: 1.2 in the version used, 3.7 in the comparative (difference of 2.4)